Continuous Microclimate Monitoring and Environmental Characterization for Preventive Conservation at Butrint National Park

READ
Research Expertise from the Academic Diaspora

READ Fellowship Program · 2026

Building a lasting environmental record

RESIDENT RESEARCH TEAM · UNIVERSITY OF TIRANA

Ervin Kafexhiu *(project leader)* · Silvana Miço · Gerti Xhixha · Mirela Alushllari · Amanda Kote · Olta Çakaj
Instrumentation · Data systems · Conservation materials

DIASPORA SCHOLAR · Arta Dilo · TNO Netherlands

Geoinformatics · Earth observation · Machine learning · Sensor networks · Mathematical optimization / Operations research

SITE PARTNER · BUTRINT MANAGEMENT FOUNDATION

Site knowledge · Conservation priorities · Field access

Resident–diaspora collaboration with BMF site expertise.

From episodic studies to lasting environmental evidence

Butrint has valuable environmental studies, but no systematic, continuous record for long-term conservation research.

01 HISTORICAL CONTEXT

Integrate existing evidence

Identify and assess relevant climate, satellite and other observations from different sources and periods.

Diaspora scholar + resident team

02 CONTINUOUS BASELINE

Establish a permanent record

Operate three reliable stations with quality-controlled microclimate measurements and durable data storage.

Instrumentation + data team

03 TARGETED INVESTIGATIONS

Study material-relevant factors

Measure moisture, salts and selected environmental or surface-condition indicators through field campaigns.

Chemistry + archaeological materials

SCIENTIFIC VALUE

An expanding evidence base for preventive conservation, future deterioration models and better monitoring decisions.

An integrated methodology, delivered by one team

Data analysis informs site selection; permanent monitoring and material investigations proceed in parallel.

01 · MONTHS 1–2

Select three informative locations

Prior studies · historical data · BMF expertise · conservation and practical constraints

02A · MONTHS 1–11

Develop & operate

Build and validate nodes, communications and database; begin continuous QA/QC observations.

Resident instrumentation and data team

02B · MONTHS 4–11

Investigate selectively

Study material moisture, salts, surface condition and complementary environmental exposures.

Chemistry and archaeological-material team

03 · THROUGHOUT THE PROJECT

Integrate evidence · transfer analytical methods · design expansion

Diaspora Scholar and resident researchers collaborate on historical-data synthesis, interpretation and training.

What READ funding leaves behind

First-phase infrastructure.

Scientific data.

Transferable capabilities.

1 **Permanent observational capability**

Three operational nodes, resilient communications and a reusable data platform.

2 **A growing environmental evidence base**

Quality-controlled continuous records, plus integrated historical sources.

3 **Targeted conservation-relevant research**

Targeted material and conservation research

Material-moisture measurements and seasonal data to identify future monitoring priorities.

4 **Collaboration and scientific capacity**

Shared methods, training, documented procedures and a joint publication target.

Long-term value: reliable model inputs, reusable infrastructure and trained researchers.

Research budget

2,471,346

ALL · TOTAL REQUESTED

A permanent, expandable monitoring foundation plus complementary research on archaeological materials.

Submitted allocation (Albanian lek)



Infrastructure 61% — **Research 39%**

INFRASTRUCTURE	1,507,000 ALL
Three-node monitoring system	1,400,000
Material moisture measurement system	107,000

RESEARCH	964,346 ALL
Resident scholar payments	764,346
Publications and others	100,000

Questions?

Three permanent monitoring points.
One lasting foundation for conservation science.